

MAT457 Lecture Notes

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1 Day 1: Here we go again! (Sep. 9, 2026)

Nabutovsky's office is in BA6104. Exams will be “difficult”, and you will be responsible for statements and proofs of theorems. You should know the meaning of the material. Unlike most Nabutovsky courses, tests will not be based on homework questions.

Undergraduate students are expected to know the basics of metric spaces, topological spaces, and set cardinality. You're also expected to have some exposure to the Lebesgue measure on $\mathbb{R}$ and $\mathbb{R}^n$, along with some of its properties.

Nabutovsky proceeds to survey the class about what they know. Here are the things he's asked (as to who is familiar on these).

1. Lebesgue density theorem in $\mathbb{R}^n$.
2. Fubini's theorem.
3. L^p spaces, and how to prove that they are complete.
4. Why care about the Lebesgue integral when the Riemann integral exists? (Per Nabutovsky, most functions you deal with in PDEs and other subjects are smooth, for which the Riemann integral will suffice. So why care?)

“Now I know where the class stands, and I will explain all these to you in due course.” In this course, we will extend the theory of Lebesgue integration to abstract spaces. Nabutovsky believes in starting slowly (with the later part of the course being much faster). We will start with the Lebesgue measure on $\mathbb{R}$.

Consider the space $X = \mathbb{R}$. Let us define a function $f: \mathcal{P}(\mathbb{R}) \rightarrow [0, +\infty]$ (observe that $+\infty$ is allowed). We want to make f a *measure*; in order to do that, it must satisfy

1. $f(\emptyset) = 0$,
2. (Countable additivity). If $E_1, E_2, \dots$ is a countable collection of disjoint sets, then

$$f\left(\bigcup_i E_i\right) = \sum_i f(E_i).$$

In particular, it follows from the above that if some E_i is of infinite measure, then $\bigcup_i E_i$ is also of infinite measure.

The following two properties apply only to the case when $X = \mathbb{R}$.

3. $f([0, 1]) = 1$,
4. f is translation-invariant, i.e., $f(E + t) = f(E)$, for all $E \in \mathcal{P}(\mathbb{R})$ and all $t \in \mathbb{R}$. Note that the set $E + t$ is given by $E + t = \{x \mid x - t \in E\}$.

At this point, Nabutovsky states that properties (iii) and (iv) are “kind of arbitrary”. Nabutovsky also said that there is in fact no function f that satisfies all 4 of these properties.

Proof. Consider the set $\mathbb{R}/\mathbb{Q}$. For convenience, let us restrict X down to $(0, 1)$; we say that $x \sim y$ if $y = x + q$, where $q \in \mathbb{Q}$, where the special addition symbol is defined by

$$a \circ b = \begin{cases} a + b & \text{if } a + b \in (0, 1), \\ a + b - 1 & \text{if } a + b \geq 1. \end{cases}$$

Let E consist of exactly one element from each of the equivalence classes of $X/\mathbb{Q}$ (from the axiom of choice)¹, for which the cardinality of E is $\mathbb{R}$. Now, define $E_q := E + q$ for $q \in \mathbb{Q} \cap [0, 1)$; then, for $q_1 \neq q_2$, we see that $E_{q_1} \cap E_{q_2} = \emptyset$, since, for all $t_1, t_2 \in E$ satisfying $t_1 + q_1 = t_2 + q_2$, it must be that

$$t_1 = t_2 + (q_2 - q_1) \implies t_1 \sim t_2,$$

so $t_1 = t_2$. Now,

$$0 < \mu((0, 1)) = \sum_{q \in \mathbb{Q} \cap [0, 1)} \mu(E_q).$$

By translation-invariance, it must be that $\mu(E_{q_1}) = \mu(E_{q_2})$, so we have two cases to consider: either $\mu(E_q) = 0$ (for any $q \in \mathbb{Q} \cap [0, 1)$), or it is greater than 0. In either case, we would get that the sum is 0 or $+\infty$, which contradicts property (iii). $\square$

At this point, Nabutovsky gave an example of another measure.

Definition 1 (Counting measure). *The counting measure counts the number of elements in a set; if there are infinitely many elements, the measure is infinite; if the set is finite, then the measure is exactly the number of elements in the set, per its definition.*

Definition 2 (Algebra). *Let X be a metric (or a topological space). Let $M \subset \mathcal{P}(X)$ (with $M \neq \emptyset$). We say M is an algebra if:*

1. *If $E \in M$, then its complement E^c is in M .*
2. *If $E_1, \dots, E_n \in M$, then their union $\bigcup_{i=1}^n E_i$ is in M .*

Definition 3 (σ -algebra). *In addition to the above properties, we ask that the union of any countable collection of elements of M is also in M .*

Using the fact that $A \cap B = (A^c \cup B^c)^c$, we also get that the countable intersection of elements of M is also in M .

¹note: this is a Vitali set

2 Day 2 - σ -algebras, measures (Sep 14, 2026)

Last time we defined the notion of a σ -algebra. Let us begin with some examples:

1. $\mathcal{M} = \{\emptyset, X\}$.
2. $\mathcal{M} = \mathcal{P}(X)$.
3. $\mathcal{M} = \{A \subseteq X : A \text{ or } A^c \text{ is countable}\}$.²

Suppose that we have some collection of subsets $A_\alpha \subseteq X$ and would like to find some σ -algebra $\mathcal{M}$ so that $A_\alpha \in \mathcal{M}$ for all α . First, observe that if $\mathcal{M}_\beta$ is some collection of σ -algebras on X and $A, A_1, A_2, \dots \in \mathcal{M}_\beta$ for all β , then for each β $A^c \in \mathcal{M}_\beta$ and $\bigcup_{j=1}^\infty A_j \in \mathcal{M}_\beta$, as $\mathcal{M}_\beta$ is a σ -algebra, and thus $\bigcap_\beta \mathcal{M}_\beta$ is also a σ -algebra. In particular, the intersection of all σ -algebras containing all A_α is also a σ -algebra containing all the A_α .³ Indeed this is also the smallest σ -algebra containing all of the A_α . We call this construction the σ -algebra generated by the A_α .

If X is a metric (or topological) space, we define the Borel σ -algebra of X to be the σ -algebra generated by the open subsets. A Borel set is an element of the Borel σ -algebra. If X is, for instance, separable,⁴ then the Borel σ -algebra is generated by the open balls.⁵ This is not the case in general. Consider $X = \mathbb{R}$ with the discrete metric $d(x, y) = 1$ if and only if $x \neq y$ and 0 otherwise. In this case, the σ -algebra generated by all open sets is clearly $\mathcal{P}(\mathbb{R})$, but the σ -algebra generated by the open balls is simply $\{\emptyset, \mathbb{R}\}$.

Let us devote some attention to the Borel σ -algebra of $\mathbb{R}$.

Proposition 4. *The Borel σ -algebra of $\mathbb{R}$ is generated by any of the following:*

1. *The set of all open intervals (a, b) .*
2. *The set of all closed intervals $[a, b]$.*
3. *The set of all half-open intervals $[a, b)$.*
4. *The set of all open rays (a, ∞) .*
5. *The set of all closed rays $(-\infty, a]$.*

Proof. $(2 \subseteq 1) [a, b] = \bigcap_{n=1}^\infty (a - 1/n, b + 1/n)$. $(3 \subseteq 2) [a, b) = \bigcup_{n=1}^\infty [a, b - 1/n]$. $(4 \subseteq 3) [a, \infty) = \bigcup_{n=1}^\infty [a, n)$ and $(a, \infty) = \bigcup_{n=1}^\infty [a + 1/n, \infty)$. $(5 \subseteq 4) (-\infty, a] = (a, \infty)^c$. $(1 \subseteq 5) (a, b] = (-\infty, b] \cap (-\infty, a]^c$. $(a, b) = \bigcup_{n=1}^\infty (a, b - 1/n]$. And so the σ -algebras generated by any of (1)-(5) are equal. Since the Borel σ -algebra of $\mathbb{R}$ is, by the above remark, generated by the open balls, we are done. $\square$

Detour: what exactly are the Borel subsets of $\mathbb{R}$? Clearly the Borel σ -algebra contains all open, closed, G_δ , and F_σ sets, but it also contains all countable intersections of F_σ sets. Let us introduce

²It is clear that $\emptyset \in \mathcal{M}$ and that $\mathcal{M}$ is closed under complementation. Suppose $A_1, \dots \in \mathcal{M}$. If all the A_j are countable then $\bigcup_{j=1}^\infty A_j$ is countable and we are done. Otherwise, A_k^c is countable for some k , but then $(\bigcup_{j=1}^\infty A_j)^c = \bigcap_{j=1}^\infty A_j^c \subseteq A_k^c$, and so is countable.

³This collection is non-empty as, for instance, $\mathcal{P}(X)$ is a σ -algebra containing all the A_α .

⁴According to Nabu, we will almost always assume this to be the case.

⁵It is easy to check that, in this case, every open set is the countable union of open balls.

some notation. Let us say that A is a Π_n set if it can be written of the form

$$\bigcap_{i_1=1}^{\infty} \bigcup_{i_2=1}^{\infty} \cdots \left(\bigcup \bigcap \right)_{i_n=1}^{\infty} A_{i_1, i_2, \dots, i_n}$$

where the $A_{i_1, i_2, \dots, i_n}$ are either open or closed, i.e. starting⁶ with an intersection then alternating between intersection and union a total of n times. Similarly, let us say that A is a Σ_n set if it can be written of the form

$$\bigcup_{i_1=1}^{\infty} \bigcap_{i_2=1}^{\infty} \cdots \left(\bigcup \bigcap \right)_{i_n=1}^{\infty} A_{i_1, i_2, \dots, i_n}$$

where again the $A_{i_1, i_2, \dots, i_n}$ are either open or closed. Is every Borel subset of $\mathbb{R}$ a Π_n or Σ_n for some n ? No! Indeed, this recipe can be generalized to arbitrary countable ordinals⁷ He then alluded to the continuum hypothesis. My take on the point we was trying to make is how we may use transfinite induction to “explicitly” determine all of the Borel subsets by iterating the above process over the countable ordinals, a procedure which “terminates at the first uncountable ordinal.”⁸ Nabu also made the observation that the cardinality of the Borel σ -algebra of $\mathbb{R}$ is the cardinality of the continuum, so that’s something.

Later, we will often prefer to work with disjoint unions as opposed to arbitrary unions. This is where “disjointification” comes in. Given $A_n \in \mathcal{M}$, we may define $B_1 = A_1$ and $B_{n+1} = A_{n+1} \cap \bigcap_{j=1}^n A_j^c$. Note that the B_m are disjoint, lie in $\mathcal{M}$, and $\bigcup_{n=1}^{\infty} A_n = \bigcup_{m=1}^{\infty} B_m$.

Suppose we have some collection of measurable spaces⁹ $(X_\alpha, \mathcal{M}_\alpha)$ and want to define a σ -algebra on the product $\prod_\alpha X_\alpha$, which we’ll denote by¹⁰ $\bigotimes_\alpha \mathcal{M}_\alpha$. To do this, we mimic the construction of the product topology in topology, and take the σ -algebra generated by the preimages $\pi_\beta^{-1}(A)$ for $A \in \mathcal{M}_\beta$, where $\pi_\beta : \prod_\alpha X_\alpha \rightarrow X_\beta$ is the projection map. For example, consider $\mathbb{R} \times \mathbb{R}$ where we take each copy of $\mathbb{R}$ to have the Borel σ -algebra. Since the open intervals generate the Borel σ -algebra of $\mathbb{R}$, it follows¹¹ that the product σ -algebra is generated by the open rectangles $(a, b) \times (c, d)$. These in turn are the open balls of $\mathbb{R}^2$ under the metric $d_{\max}(x, y) = \|x - y\|_\infty = \max(|x_1 - y_1|, |x_2 - y_2|)$; since all norms on $\mathbb{R}^2$ are equivalent, the product σ -algebra is also generated by the usual open balls of $\mathbb{R}^2$, i.e. the product σ -algebra of $\mathbb{R} \times \mathbb{R}$ and the Borel σ -algebra of $\mathbb{R}^2$ coincide.

Definition 5 (Measure). *Let $(X, \mathcal{M})$ be a measurable space. A measure is a map $\mu : \mathcal{M} \rightarrow [0, \infty]$ such that:*

1. $\mu(\emptyset) = 0$.
2. “Countable additivity” If $A_n \in \mathcal{M}$ are disjoint then $\mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_n)$.

Let’s see some examples:

1. $\mathcal{M} = \mathcal{P}(X)$ and $\mu(A) = \infty$ if A has infinitely many elements and $\mu(A) = |A|$ (the cardinality of A) otherwise. More generally, given $f : X \rightarrow [0, \infty]$ we may define $\mu_f(A) = \sum_{a \in A} f(a)$.
2. $\mathcal{M} = \mathcal{P}(X)$ and $\mu(A) = 0$ if A is countable and ∞ otherwise.

⁶Reading left-to-right.

⁷An ordinal is an equivalence class of ordered sets.

⁸Someone once told me that transfinite induction was first used/formulated to “describe” the Borel subsets of $\mathbb{R}$; so be sure to memorize that for pub trivia.

⁹A measurable space is a non-empty set X equipped with a σ -algebra $\mathcal{M}$.

¹⁰:(.

¹¹Exercise!

3. Given a measure $\mu_0 : \mathcal{M}_0 \rightarrow [0, \infty]$ and $A \subseteq X$, define $\mathcal{M} = \{B \subseteq X : A \cap B \in \mathcal{M}_0\}$ and $\mu : \mathcal{M} \rightarrow [0, \infty]$ by $\mu(B) = \mu_0(A \cap B)$. “The restriction measure”
4. Given $x_0 \in X$, define $\mu(A) = 1$ if $x_0 \in A$ and 0 otherwise. “Dirac measure”
5. Lebesgue measure on $\mathbb{R}^n$.

The following is a non-example. Define $\mu(A) = \infty$ if A is infinite and 0 otherwise.

Definition 6 (Finite). *A measure μ is finite if $\mu(X) < \infty$.*

For example, $\mu(A) = \int_A e^{-x^2} dx$ on the σ -algebra of Lebesgue measurable subsets of $\mathbb{R}$, the Dirac measure. A non-example: the Lebesgue measure.

Definition 7 (σ -finite). *A measure μ is σ -finite if there are countably many measurable sets E_i so that $\mu(E_i) < \infty$ and $X = \bigcup_{i=1}^{\infty} E_i$.*

For example, the Lebesgue measure. A non-example: the counting measure on $\mathbb{R}$.

Definition 8 (Semi-finite). *A measure μ is semi-finite if $E \in \mathcal{M}$ and $\mu(E) = \infty$ implies there is some $F \subseteq E$ with $F \in \mathcal{M}$ and $0 < \mu(F) < \infty$.*

Exercise 9. *Show that σ -finite implies semi-finite, but that the converse may fail.*

Definition 10 (Measure space). *A measure space is a triple $(X, \mathcal{M}, \mu)$ where X is a non-empty set, $\mathcal{M}$ is a σ -algebra on X , and $\mu : \mathcal{M} \rightarrow [0, \infty]$ is a measure.*

Theorem 11. *Let $(X, \mathcal{M}, \mu)$ be a measure space. We have:*

1. “Monotonicity” If $A, B \in \mathcal{M}$ and $A \subseteq B$ then $\mu(A) \leq \mu(B)$.
2. “Countable subadditivity” If $A_n \in \mathcal{M}$ then $\mu(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mu(A_n)$.
3. “Continuity from below” If $E_1 \subseteq E_2 \subseteq \dots$ then $\mu(\bigcup_{n=1}^{\infty} E_n) = \lim_{n \rightarrow \infty} \mu(E_n)$.
4. “Continuity from above” If $E_1 \supseteq E_2 \supseteq \dots$, $\mu(E_1) < \infty$ then $\mu(\bigcap_{n=1}^{\infty} E_n) = \lim_{n \rightarrow \infty} \mu(E_n)$.

Proof. For (1), note that B may be written as the disjoint union of A and $B \cap A^c$, so by additivity $\mu(B) = \mu(A) + \mu(B \cap A^c) \geq \mu(A)$, since μ is non-negative.¹² For (2), apply disjointification, countable additivity, and monotonicity to obtain $\mu(\bigcup_{n=1}^{\infty} A_n) = \mu(\bigcup_{m=1}^{\infty} B_m) = \sum_{m=1}^{\infty} \mu(B_m) = \sum_{m=1}^{\infty} \mu(A_m \cap \bigcap_{j=1}^{m-1} A_j) \leq \sum_{m=1}^{\infty} \mu(A_m)$. For (3), take $B_1 = E_1$ and $B_{m+1} = E_{m+1} \setminus E_m$. Since the B_m are disjoint and $\bigcup_{n=1}^{\infty} E_n = \bigcup_{m=1}^{\infty} B_m$, we have $\mu(\bigcup_{n=1}^{\infty} E_n) = \mu(\bigcup_{m=1}^{\infty} B_m) = \sum_{m=1}^N \mu(B_m) + \sum_{m=N+1}^{\infty} \mu(B_m) = \mu(\bigcup_{m=1}^N B_m) + \sum_{m=N+1}^{\infty} \mu(B_m) = \mu(E_N) + \sum_{m=N+1}^{\infty} \mu(B_m)$, where the latter summand is the tail of a convergent series, and so vanishes in the limit. For (4), put $F_j = E_1 \setminus E_j$. Notice that $F_1 \subseteq F_2 \subseteq \dots$ and so by (3) we have $\mu(\bigcup_{j=1}^{\infty} F_j) = \lim_{j \rightarrow \infty} \mu(F_j)$. Now, $\mu(F_j) = \mu(E_1) - \mu(E_j)$ by additivity; and $\bigcup_{j=1}^{\infty} F_j = E_1 \setminus \bigcap_{n=1}^{\infty} E_n$. All together, we have $\mu(E_1) - \mu(\bigcap_{n=1}^{\infty} E_n) = \mu(\bigcup_{j=1}^{\infty} F_j) = \mu(E_1) - \lim_{n \rightarrow \infty} \mu(E_n)$. Since $\mu(E_1) < \infty$, we conclude. $\square$

Note that the hypothesis in (4) that $\mu(E_1) < \infty$ is important. Consider the Lebesgue measure on $\mathbb{R}$ and $E_i = (i, \infty)$. Then $\bigcap_{i=1}^{\infty} E_i = \emptyset$ but $\lim_{i \rightarrow \infty} \mu(E_i) = \infty$.

A measurable set of measure zero is called a null set. By monotonicity, measurable subsets of null sets are null sets. However, it could be the case that a subset of a null set fails to be measurable. For a trivial example, take $X = \{a, b\}$, $\mathcal{M} = \{\emptyset, X\}$, and μ the counting measure.

¹²Later in the course, we will discuss signed measures, which are precisely what it says on the tin.

Definition 12 (Complete measure). *A measure is said to be complete¹³ if all subsets of null sets are measurable.*

Definition 13 (Completion). *Let $(X, \mathcal{M}, \mu)$ be a measure space. Let $\overline{\mathcal{M}}$ be the σ -algebra generated by $\mathcal{M}$ and the subsets of null sets. The (unique) extension of μ to $\overline{\mathcal{M}}$ is called the completion.*

Nabu claimed that $\overline{\mathcal{M}} = \{A \cup N : A \in \mathcal{M}, N \subseteq Z \in \mathcal{M} \text{ with } \mu(Z) = 0\}$. He didn't have time for a proof, but I've provided one below. While technically it should be a theorem that the extension of a measure actually exists, this result makes it clear that setting $\overline{\mu}(A \cup N) = \mu(A)$ is well-defined. Uniqueness, along with some other checks, are straight-forward.¹⁴

Proposition 14. $\overline{\mathcal{M}} = \{A \cup N : A \in \mathcal{M}, N \subseteq Z \in \mathcal{M} \text{ with } \mu(Z) = 0\}$.

Proof. $\emptyset$ is a null set, so $\emptyset = \emptyset \cup \emptyset \in \overline{\mathcal{M}}$. By countable subadditivity, the countable union of null sets is a null set, taken with the identity $\bigcup_{n=1}^{\infty} A_n \cup N_n = (\bigcup_{n=1}^{\infty} A_n) \cup (\bigcup_{n=1}^{\infty} N_n)$, it quickly follows that $\overline{\mathcal{M}}$ is closed under countable unions. Now suppose $E = A \cup N \in \overline{\mathcal{M}}$ and pick $Z \in \mathcal{M}$ with $N \subseteq Z$ and $\mu(Z) = 0$. Now, $N \subseteq Z$ implies $Z^c \subseteq N^c$ and so $N^c = Z^c \cup (Z \setminus N)$. Substituting this in, we have $E^c = A^c \cap N^c = (A^c \cap Z^c) \cup (A^c \cap (Z \setminus N))$. But $A^c, Z^c \in \mathcal{M}$ and $A^c \cap (Z \setminus N) \subseteq Z$ which has measure zero, and thus E^c is of the desired form. $\square$

¹³This terminology has nothing to do with the notion of completeness for metric spaces.

¹⁴Exercise!

3 Day 3 - Outer measures

Last time: we talked about complete measures and completions of measures. We started with the proposition from last time about the σ -algebra generated by $\mathcal{M}$ and the subsets of null sets being all those sets of the form $M \cup N$ for $M \in \mathcal{M}$ and $N \subseteq Z$ for some $Z \in \mathcal{M}$ with $\mu(Z) = 0$. The proof Nabu gave in lecture is essentially the one I gave in Day 2. Nabu also spelled out why the extension $\bar{\mu}(M \cup N) = \mu(M)$ is well-defined. Suppose that $M_1 \cup N_1 = M_2 \cup N_2$ for M_i, N_i , and $N_i \subseteq Z_i$ as before; we need to show that $\mu(M_1) = \mu(M_2)$. By symmetry, it suffices to show that $\mu(M_1) \leq \mu(M_2)$. Now, $M_1 \subseteq M_1 \cup N_1 = M_2 \cup N_2 \subseteq M_2 \cup Z_2$ and so by monotonicity and subadditivity, $\mu(M_1) \leq \mu(M_2) + \mu(Z_2) = \mu(M_2)$, and we are done.

We now move our attention to so-called “outer measures.”

Definition 15 (Outer measure). *A function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ is an outer measure if:*¹⁵

1. $\mu^*(\emptyset) = 0$.
2. μ^* has monotonicity.
3. μ^* is countably subadditive.

Differences: “no pesky σ -algebras;” measures have monotonicity and countable subadditivity, but also countable additivity (on disjoint sets). Question: “What’s so outer about this measure?”

To answer this, let’s consider an example on $X = \mathbb{R}$. Let us start by setting the outer measure of an interval¹⁶ to be its length. Then define $\mu^*(A) = \inf\{\sum_{i=1}^{\infty} (b_i - a_i) : \bigcup_{i=1}^{\infty} (a_i, b_i) \supseteq A\}$.¹⁷ Recall that this is the first step in defining the Lebesgue measure.

Question: do we really need countable collections? Yes, since there is no finite collection of intervals which cover $\mathbb{R}$. What if we restrict to bounded sets? The set $\mathbb{Q} \cap [0, 1]$ is bounded and has outer measure 0, as it should, but if we restrict to finite collections, then the sums of lengths of intervals of any finite covering of $\mathbb{Q} \cap [0, 1]$ is at least 1,¹⁸ and so this notion of outer measure would not play well with limits, which messes with the completeness of spaces of integrable functions.

We can mimic this construction in much more generality.

Theorem 16. *Suppose that $\mathcal{E} \subseteq \mathcal{P}(X)$ such that $\emptyset, X \in \mathcal{E}$ and that $\lambda : \mathcal{E} \rightarrow [0, \infty]$ satisfies $\lambda(\emptyset) = 0$. Define $\mu^*(A) = \inf\{\sum_{i=1}^{\infty} \lambda(E_i) : \bigcup_{i=1}^{\infty} E_i \supseteq A, E_i \in \mathcal{E}\}$. Then, μ^* is an outer measure.*

Proof. Note that μ^* is well-defined since $\{X\} \subseteq \mathcal{E}$ is a countable cover of any subset $A \subseteq X$. If $A \subseteq B$ then every covering of B is also a covering of A , and so $\mu^*(A) \leq \mu^*(B)$ since the infimum over a larger set is at most the infimum over a smaller set. Now, fix $\varepsilon > 0$ and suppose $A_i \subseteq X$. Fix i , by definition, we may pick $E_{i,j} \in \mathcal{E}$ so that $\bigcup_{j=1}^{\infty} E_{i,j} \supseteq A_i$ and $\sum_{j=1}^{\infty} \lambda(E_{i,j}) \leq \mu^*(A_i) + 2^{-i}\varepsilon$. Then $\{E_{i,j}\}$ is a countable covering of $\bigcup_{i=1}^{\infty} A_i$ by sets in $\mathcal{E}$, and so $\mu^*(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \lambda(E_{i,j}) \leq \sum_{i=1}^{\infty} \mu^*(A_i) + 2^{-i}\varepsilon = \varepsilon + \sum_{i=1}^{\infty} \mu^*(A_i)$. Since ε was arbitrary, we are done. $\square$

¹⁵The notions of monotonicity and countable subadditivity coincide with their use in **Theorem 11**

¹⁶Of any kind.

¹⁷In this example we’ve required the covering to consist of open intervals, but nothing changes by replacing these with some other interval.

¹⁸Convince yourself, using topology, that if finitely many open intervals cover $\mathbb{Q} \cap [0, 1]$ then they must also cover $[0, 1]$.

So it is very easy to construct outer measures. Next, much like how the above theorem generalizes the outer measure of $\mathbb{R}$, we can also mimic the construction of the Lebesgue measure from the outer measure at the level of generality of the above.

Definition 17 (μ^* -measurable). *Given an outer measure μ^* , we say a set $A \subseteq X$ is μ^* -measurable if for all $E \subseteq X$, $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$.*

And, as in the case of the Lebesgue measure, this procedure produces a (complete) measure.

Theorem 18 (Caratheodory). *If μ^* is an outer measure on X , then*

1. *The collection of μ^* -measurable sets, which we'll denote by $\mathcal{M}$, is a σ -algebra.*
2. *The restriction of μ^* to $\mathcal{M}$ is a complete measure.*

We shall delay the proof for now. Let us discuss what we don't get from Caratheodory's theorem. We don't get any real description of $\mathcal{M}$ in this theorem, a priori it could be that $\mathcal{M} = \{\emptyset, X\}$. We are also not guaranteed that the restriction of μ^* to $\mathcal{E} \cap \mathcal{M}$ should agree with λ . "In such generality we don't often get what we want."

Let us argue why the restriction of the outer measure to intervals does agree with length. Without loss of generality, we may assume the coverings in the definition of the outer measure consist of open intervals. Similarly, it suffices to show the claim for closed intervals.¹⁹ Fix a countable cover $I_i = (a_i, b_i)$ of $[a, b]$. By compactness, there exists K so that $[a, b] \subseteq \bigcup_{i=1}^K I_i$. Fix a large number N and consider the scaled intervals (Na_i, Nb_i) and $[Na, Nb]$. Let S be the number of integral lattice points in $[Na, Nb]$ and let S_i be the number of integral lattice points in (Na_i, Nb_i) . By the standard bounds for the floor and ceiling functions, it follows that $S_i \leq N(b_i - a_i) + 1$ and $S \geq N(b - a) - 1$. Since $I_1, \dots, I_K$ cover $[a, b]$, we have $S \leq \sum_{i=1}^K S_i$. All together,

$$b - a - \frac{1}{N} \leq \frac{S}{N} \leq \sum_{i=1}^K \frac{S_i}{N} \leq \frac{K}{N} + \sum_{i=1}^K (b_i - a_i) \leq \frac{K}{N} + \sum_{i=1}^{\infty} (b_i - a_i)$$

Taking N to infinity, we obtain $b - a \leq \sum_{i=1}^{\infty} (b_i - a_i)$ and so $b - a \leq \mu^*([a, b])$. The other inequality is clear since $\{[a, b]\}$ is a countable cover of $[a, b]$ with total length $b - a$. $\square$

¹⁹Due to time reasons, Nabu couldn't finish the proof of this claim. What follows this footnote is my fleshing out of what he vibe-coded, and therefore could be completely different from what he had in mind.

4 Day 4 - TA Guest Lecture! and Caratheodory's theorem

Today's lecture was given our TA.²⁰ Most of the lecture will be guided by the question: "is there a Borel measure?" Of course, we will answer this in much more generality.

Definition 19 (Premeasure). *Let $\mathcal{A}$ be an algebra.²¹ A premeasure is a map $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ satisfying:*

1. $\mu_0(\emptyset) = 0$.
2. *Countable additivity (for disjoint unions).*

Roughly speaking, we will follow these steps: (1) define a premeasure on $\mathcal{A}$,²² (2) get an outer measure from this premeasure, (3) get a σ -algebra $\mathcal{M}$ from this outer measure so that the restriction to $\mathcal{M}$ is a complete measure, and (4) show that this complete measure is "almost" unique.

Let $\mathcal{A}$ be the smallest algebra containing all intervals of the form $(a, b]$, $a < b$. Note that since $\mathcal{A}$ is only an algebra, it is distinct from, say, the smallest algebra containing all open intervals.

Proposition 20. *$E \in \mathcal{A}$ if and only if $E = \bigcup_{j=1}^n (a_j, b_j]$.*

Proof. It suffices to show that the sets of this form are an algebra. Taking $n = 0$ yields $\emptyset$. Closure under finite unions is clear. For complementation, observe that $(a_j, b_j]^c = (-\infty, a_j] \cup (b_j, \infty)$, and that for a general union we can "fill in the gaps" using intervals of the correct form. $\square$

From **Proposition 20**, it is clear that defining $\mu_0(E) = \sum_{j=1}^n b_j - a_j$ gives a premeasure. This completes step 1 for the construction of the Borel measure.

Proposition 21. *Let $\mathcal{A}$ be an algebra and μ_0 a premeasure on $\mathcal{A}$. Define, as before, for $E \subseteq X$, $\mu^*(E) = \inf\{\sum_{j=1}^{\infty} \mu_0(A_j) : A_j \in \mathcal{A}, E \subseteq \bigcup_{j=1}^{\infty} A_j\}$. Then, μ^* is an outer measure. Further, μ^* extends μ_0 .*

Proof. That μ^* is an outer measure was seen before as **Theorem 16**. Since $\{E\}$ is always a countable cover of E , it is clear that $\mu_0(E) \geq \mu^*(E)$. For the other inequality, notice it suffices to show that $\mu_0(E) \leq \sum_{j=1}^{\infty} \mu_0(A_j)$ for any countable cover $A_j \in \mathcal{A}$ of E . But this follows from monotonicity and then countable subadditivity.²³ $\square$

The above proof should be contrasted with the one Nabu tried to give for the Lebesgue measure. In particular, starting with a premeasure on $\mathcal{A}$ instead of only having λ of intervals helps a lot. This wraps up step 2.

The jump from an outer measure to a suitable σ -algebra is called Caratheodory's theorem. Recall that we say $A \subseteq X$ is μ^* -measurable if for all $E \subseteq X$ we have $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$.

Proof of Caratheodory's theorem (Theorem 18). Let $\mathcal{M}$ be all those sets which are μ^* -measurable. On Day 3 we showed that $\emptyset \in \mathcal{M}$, $\mathcal{M}$ is closed under complementation, and that $\mu^*(\emptyset) = 0$. Let

²⁰Or, perhaps, Nabu after very extensive plastic surgery.

²¹Like a σ -algebra, except we only require closure under finite unions.

²²This is the only place we will use the specifics that lead to the Borel measure.

²³Essentially the same argument that a measure is countably subadditive also demonstrates that a premeasure is countably subadditive.

us show that $\mathcal{M}$ is closed under finite unions and that $\mu^*|_{\mathcal{M}}$ is finitely additive. For $A, B \in \mathcal{M}$, $E \subseteq X$, we have:

$$\begin{aligned}\mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &= \mu^*((E \cap A) \cap B) + \mu^*((E \cap A) \cap B^c) + \mu^*((E \cap A^c) \cap B) + \mu^*((E \cap A^c) \cap B^c) \\ &\geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c) \\ &\geq \mu^*(E)\end{aligned}$$

And so $A \cup B \in \mathcal{M}$. Applying this to $E = A$ in the case where A, B are disjoint, we have $\mu^*(A \cup B) = \mu^*((A \cup B) \cap A) + \mu^*((A \cup B) \cap A^c) = \mu^*(A) + \mu^*(B)$. And so we have finite additivity. Let us now show that $\mathcal{M}$ is closed under countable unions and that $\mu^*|_{\mathcal{M}}$ is countably additive. By disjointification, we may assume without loss of generality that the $A_j \in \mathcal{M}$ are disjoint. Define $B_n = \bigcup_{j=1}^n A_j$, $B = \bigcup_{j=1}^{\infty} A_j$. Let $E \subseteq X$. He left as an exercise that $\mu^*(E \cap B_n) = \sum_{j=1}^n \mu^*(E \cap A_j)$. By the finite case, $B_n \in \mathcal{M}$ and so $\mu^*(E) = \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c) \geq \sum_{j=1}^n \mu^*(E \cap A_j) + \mu^*(E \cap B^c)$. Taking $n \rightarrow \infty$, we have

$$\begin{aligned}\mu^*(E) &\geq \sum_{j=1}^{\infty} \mu^*(E \cap A_j) + \mu^*(E \cap B^c) \\ &\geq \mu^*\left(E \cap \bigcup_{j=1}^{\infty} A_j\right) + \mu^*(E \cap B^c) \\ &\geq \mu^*(E)\end{aligned}$$

And so $B = \bigcup_{j=1}^{\infty} A_j \in \mathcal{M}$. If the A_j are disjoint, the trick from before works to show countable additivity. It remains to show that μ^* is complete. It suffices to show that $\mu^*(A) = 0$ implies $A \in \mathcal{M}$. But this is clear from subadditivity and monotonicity: $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c) \leq \mu^*(A) + \mu^*(E) = \mu^*(E)$. $\square$

For the full step 3, we still need to show that $\mathcal{A} \subseteq \mathcal{M}$.

Proposition 22. *If $A \in \mathcal{A}$ then $A \in \mathcal{M}$.*

Proof. Let $\varepsilon > 0$, $E \subseteq X$. Pick $B_j \in \mathcal{A}$ so that $E \subseteq \bigcup_{j=1}^{\infty} B_j$ and $\sum_{j=1}^{\infty} \mu_0(B_j) \leq \mu^*(E) + \varepsilon$.²⁴ Then,

$$\begin{aligned}\mu^*(E) + \varepsilon &\geq \sum_{j=1}^{\infty} \mu_0(B_j) \\ &= \sum_{j=1}^{\infty} \mu_0(B_j \cap A) + \sum_{j=1}^{\infty} \mu_0(B_j \cap A^c), \text{ note}^{25} \\ &= \sum_{j=1}^{\infty} \mu^*(B_j \cap A) + \sum_{j=1}^{\infty} \mu^*(B_j \cap A^c) \\ &\geq \mu^*(E \cap A) + \mu(E \cap A^c)\end{aligned}$$

and so taking $\varepsilon \rightarrow 0$ finishes the proof. $\square$

²⁴These exist by the definition of the infimum.

Let us now characterize what we mean by “almost” unique in step 4.

Proposition 23. *Let ν be another measure on $\mathcal{M}$ such that $\nu|_{\mathcal{M}} = \mu_0$. Then,*

1. $\nu(E) \leq \mu(E)$ for all $E \in \mathcal{M}$. [here $\mu = \mu^*|_{\mathcal{M}}$]
2. If $\mu(E) < \infty$ then $\nu(E) = \mu(E)$.

In particular, if μ is σ -finite then we have uniqueness.

Proof. For (1), it suffices to show that for $E \subseteq \bigcup_{j=1}^{\infty} A_j$ with $A_j \in \mathcal{A} \subseteq \mathcal{M}$, that we have the inequality $\nu(E) \leq \sum_{j=1}^{\infty} \mu_0(A_j)$. We check: $\nu(E) = \nu(\bigcup_{j=1}^{\infty} E \cap A_j) \leq \sum_{j=1}^{\infty} \nu(E \cap A_j) \leq \sum_{j=1}^{\infty} \nu(A_j) = \sum_{j=1}^{\infty} \mu_0(A_j)$. For (2), suppose $\mu(E) < \infty$ and let $\varepsilon > 0$. Pick $A_j \in \mathcal{A}$ (without loss of generality, assume they are disjoint) so that $E \subseteq \bigcup_{j=1}^{\infty} A_j$ and $\sum_{j=1}^{\infty} \mu_0(A_j) \leq \mu(E) + \varepsilon$. Let $A = \bigcup_{j=1}^{\infty} A_j \in \mathcal{M}$. Note that $\mu(A) = \nu(A)$. Since $\mu(E) < \infty$, we also have $\mu(A \setminus E) \leq \varepsilon$. Then,

$$\mu(E) \leq \mu(A) = \nu(A) = \nu(E) + \nu(A \setminus E) \leq \nu(E) + \varepsilon$$

So taking $\varepsilon \rightarrow 0$ finishes the proof. □

Now, recall that we defined μ_0 on $E = \bigcup_{j=1}^n (a_j, b_j]$ to be $\sum_{j=1}^n b_j - a_j$. But what actually was special about the length that allowed us to define a premeasure? Nothing! Indeed,

Proposition 24 (1.15 in Folland). *If $F : \mathbb{R} \rightarrow \mathbb{R}$ is increasing and right continuous, then $\mu_0\left(\bigcup_{j=1}^n (a_j, b_j]\right) = F(b_j) - F(a_j)$ is a premeasure on $\mathcal{A}$.*

We can then repeat the construction we did for the case $F(x) = x$ to obtain a measure, which we'll denote by μ_F , on $\mathcal{B}(\mathbb{R})$. We then have the following propositions about this construction.

Proposition 25 (1.16, Line 2 in Folland). *If F, G are increasing and right continuous then $\mu_F = \mu_G$ if and only if $F - G$ is constant.*

So this extension is essentially unique.

Proposition 26 (1.16, Line 3 in Folland). *Suppose μ is a measure on some $\mathcal{M}$ such that $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{M}$ with $\mu(E) < \infty$ on bounded Borel subsets $E \subseteq \mathbb{R}$. Define $F : \mathbb{R} \rightarrow \mathbb{R}$ by*

$$F(x) = \begin{cases} \mu((0, x]) & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -\mu((-x, 0]) & \text{if } x < 0 \end{cases}$$

Then F is increasing and right continuous and $\mu = \mu_F$.

Remark 27. *This construction is very similar to that of a cumulative distribution function from probability theory.*

Definition 28. *The completion of μ_F is said to be Lebesgue-Stieltjes.*

5 Day 5 - More on Caratheodory

Brief recap: Caratheodory's theorem and, in particular, the construction of the Lebesgue measure on $\mathbb{R}$. Fix $\mu : \mathcal{E} \rightarrow [0, \infty]$ where $\emptyset, X \in \mathcal{E}$ such that: $\mu(\emptyset) = 0$ and μ satisfies countable additivity. We then defined how to use this to construct an outer measure μ^* , i.e. a map $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ satisfies countably subadditivity and monotonicity. The idea of Caratheodory is then to restrict to those sets E with the property that $\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c)$. Remarkably, these sets form a σ -algebra and the restriction of μ^* to said σ -algebra is a complete measure. Note that Caratheodory's theorem²⁶ only requires μ^* to be an outer measure, it need not be that μ^* is induced by some premeasure.

In practice. Let $f(A) = 1$ if $A \neq \emptyset$ and 0 if $A = \emptyset$. Let's trace through the steps to see what measure we get from this. This function is not a premeasure. This is more in line with what Nabu did with λ in the case of the Lebesgue measure. Recall that this way of doing it was easier to define out of the gate, or rather there was less to check out of the gate, but proving that the outer measure extended λ required an argument. Let $\mu^*(A) = \inf\{\sum_{i=1}^{\infty} f(E_i) : \dots\}$. Let us check that this is an outer measure. Monotonicity is baked in. It is also clear that $\mu^*(\emptyset) = 0$ and that μ^* is countably subadditive. As a result, we can apply Caratheodory to this (we can't get that the resulting measure is an extension of f , as f is terrible). For simplicity, let's suppose $X = \mathbb{R}$. What are the μ^* -measurable sets? Suppose $E \neq \emptyset, \mathbb{R}$, then $1 = \mu^*(\mathbb{R}) \neq \mu^*(E) + \mu^*(E^c)$ since E, E^c are non-empty. Thus, the σ -algebra of μ^* -measurable sets is trivial. So this is disappointing on two fronts: the restriction of μ^* not an extension and no interesting measurable sets. This should be contrasted with the case from Day 4 that f was a premeasure, where we do get an extension of f and we do have that every element in the domain algebra of f is measurable.

He then recapped that, if we have an increasing and right continuous function $F : \mathbb{R} \rightarrow \mathbb{R}$, we can define a premeasure $\mu_0(\bigcup_{j=1}^n (a_j, b_j]) = \sum_{j=1}^n F(b_j) - F(a_j)$ on the algebra generated by the intervals of the form $(a, b]$.²⁷ And that we (essentially) have uniqueness.

Let's "remember the other story" of constructing the Lebesgue measure. We've shown that the outer measure defined in terms of λ agrees with λ on compact intervals.²⁸ We can generalize this argument to Lebesgue-Stieltjes by counting integral lattice points in $(F(a_j), F(b_j))$. Notice that this argument very much uses compactness, as the argument relies on a finite cover. In the argument on Day 3, I stated that we could assume without loss of generality that the coverings are by open intervals, the reason for this is that we can replace $(a_j, b_j]$ with (a_j, b_j) and $(b_j - 2^{-j}\varepsilon, b_j + 2^{-j}\varepsilon)$ without changing the fact that the cover is countable or the sum of the lengths modulo ε . We have not shown, however, that these sets are measurable!²⁹ Let $\varepsilon > 0$. Fix $A \subseteq \mathbb{R}$. Let $A_1 = A \cap (a, \infty)$ and let $A_2 = A \cap (-\infty, a]$. Pick a cover I_j of A by intervals so that $\sum_{j=1}^{\infty} \lambda(I_j) \leq \mu^*(A) + \varepsilon$. Let $J_j = I_j \cap (a, \infty)$ and $J'_j = I_j \cap (-\infty, a]$. Note that $\lambda(I_j) = \lambda(J_j) + \lambda(J'_j)$. Then the J_j (J'_j resp.) are a countable cover of A_1 (A_2 resp.) by intervals. So $\mu^*(A_1) + \mu^*(A_2) \leq \sum_{j=1}^{\infty} \lambda(J_j) + \sum_{j=1}^{\infty} \lambda(J'_j) = \sum_{j=1}^{\infty} \lambda(I_j) \leq \mu^*(A) + \varepsilon$. This yields the claim, as the open rays generate the Borel σ -algebra.

Next question (to think about at home): what happens when we do this slash how can we do this to an arbitrary metric space?

²⁶He's referring to the formulation of Caratheodory's theorem from Day 3. As we will see, we lose pieces of the formulation we proved on Day 4 when we have less information.

²⁷"Not very instructive, but nonetheless you will have to study this." - Nabu

²⁸In lecture, he retold the argument, as he technically left it unfinished on Day 3. But the argument he gave is essentially what I filled in on Day 3, so see that for details.

²⁹We had one minute left at this point, so the following argument is again my fleshing out of his vibe-coding.